SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS: **Total Marks: 20**

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I S.Ganesh Reddy(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

praveen
Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding

Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

ANSWERS

GITUB LINK: https://github.com/venkatapraveen1/INTERNET-PROGRAMMING

Question 1

Responsive Web Design (RWD) is a web design approach that enables a website to adapt automatically to different screen sizes such as mobile phones, tablets, and desktop computers. It ensures that the layout, images, and content are displayed properly on all devices, providing a consistent and user-friendly experience.

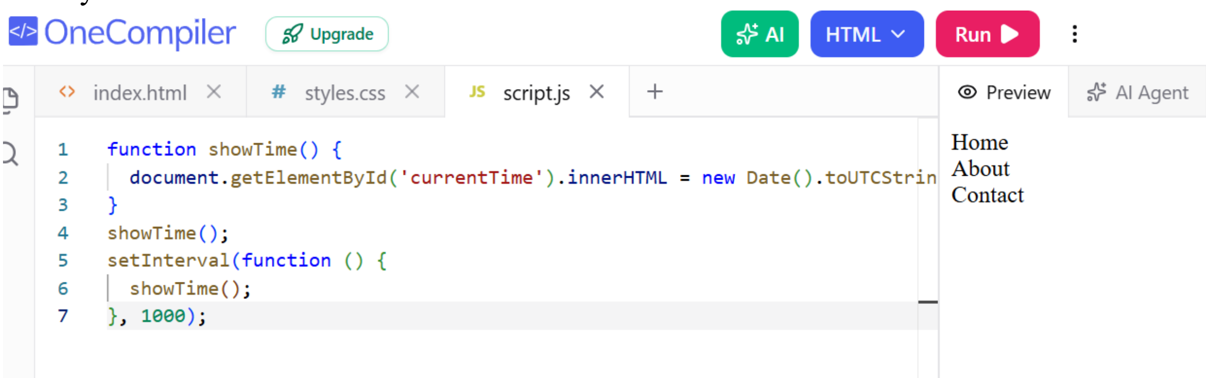
The importance of responsive web design includes improved user experience, better accessibility across devices, reduced maintenance by using a single website for all platforms, improved search engine optimization (SEO), and faster page loading.

The key CSS techniques used to achieve responsiveness are fluid layouts using percentage-based widths, flexible images with max-width: 100%, relative units such as em, rem, vw, and vh, and media queries that apply different styles based on screen size.

CSS Flexbox and CSS Grid are commonly used to create adaptive layouts. Flexbox is suitable for arranging elements in one dimension (row or column), while CSS Grid is useful for creating complex two-dimensional layouts with rows and columns. These layouts automatically adjust according to the available screen space.

A simple example is a webpage containing a navigation bar, content section, and sidebar. On a desktop, these sections appear side by side, while on a mobile device they are arranged vertically for better readability.

To improve user experience, responsive images should be used, touch-friendly buttons should be provided, font sizes should be adjusted for readability, page loading should be optimized, and layouts should be tested on different devices.



Question 2

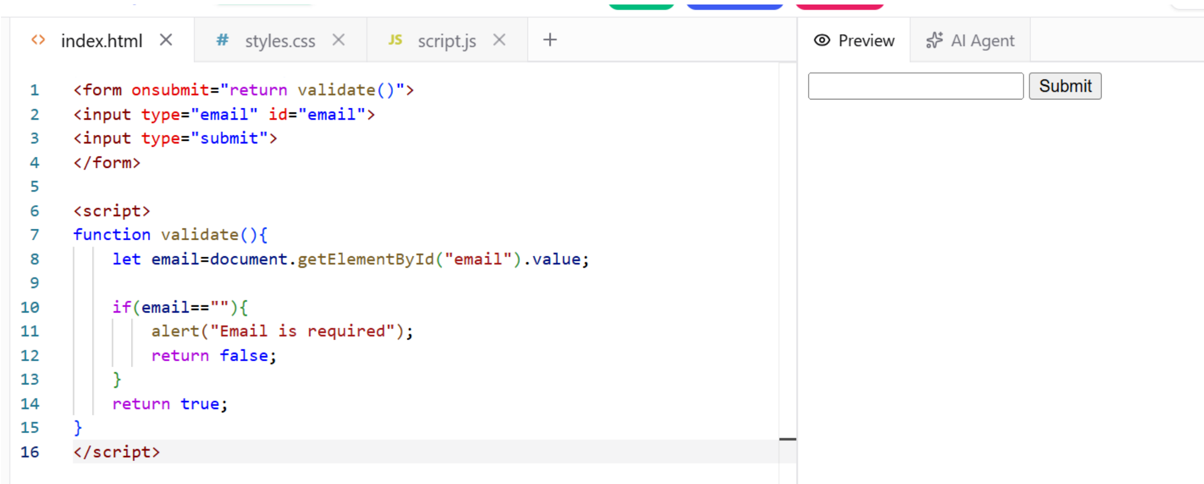
Client-side validation is the process of checking user input in the web browser before the data is sent to the server. Its main purpose is to ensure that users enter valid information, reduce unnecessary server requests, improve user experience, and provide instant feedback when incorrect data is entered.

JavaScript event handling is used to perform validation when specific events occur, such as clicking the submit button, leaving an input field, or typing in a text box. Common events include submit, blur, change, and keyup. These events trigger JavaScript functions that verify the entered data.

Regular expressions (Regex) are patterns used to validate user inputs such as email addresses, phone numbers, passwords, and other formatted data. For example, an email address must contain a username, the @ symbol, and a valid domain, while a phone number may be restricted to exactly ten digits.

For example, when a user enters an invalid email address and clicks the submit button, JavaScript checks the input using a regular expression. If the format is incorrect, an error message is displayed and the form submission is prevented until the user enters valid information.

To improve usability and security, clear error messages should be displayed near the input fields, HTML5 validation attributes should be used, both client-side and server-side validation should be implemented, strong password validation should be enforced, and user-friendly feedback should be provided.




```html
<form onsubmit="return validate()">
<input type="email" id="email">
<input type="submit">
</form>

<script>
function validate(){
    let email=document.getElementById("email").value;

    if(email==""){
        alert("Email is required");
        return false;
    }
    return true;
}
</script>
```

Question 3

The CSS Box Model is a fundamental concept that describes how every HTML element is displayed on a webpage. It consists of four components: content, padding, border, and margin. The content is the actual text or image, padding is the space around the content, the border surrounds the padding, and the margin creates space outside the border.

CSS positioning techniques determine how elements are placed on a webpage. Static positioning is the default, relative positioning moves an element relative to its original position, absolute positioning places an element relative to its nearest positioned ancestor, fixed positioning keeps an element fixed on the screen during scrolling, and sticky positioning behaves as both relative and fixed depending on the scroll position.

A webpage layout may break when the browser window is resized due to fixed-width elements, large images, incorrect positioning, overflowing content, or the absence of responsive CSS.

Media queries solve these issues by applying different CSS styles based on the screen size. They allow layouts to change from multiple columns on larger screens to a single-column layout on smaller devices, ensuring proper display and readability.

Best practices include using responsive layouts with Flexbox or Grid, avoiding fixed-width elements, using relative units, optimizing images, applying media queries at suitable breakpoints, and testing the website on different devices and browsers.

index.html × # styles.css × JS script.js × + | Preview | AI Agent

```css
@media (max-width:768px){
.container{
    flex-direction:column;
}
}
```

Submit

Question 4

The Document Object Model (DOM) is a tree-like representation of an HTML document that allows JavaScript to access, modify, add, or remove webpage elements dynamically. Every HTML element is treated as an object, making it possible to update webpage content without reloading the entire page.

The DOM plays an important role in web development by enabling dynamic content updates, modifying styles and attributes, handling user interactions, creating new elements, and removing existing elements. This allows developers to build interactive and responsive web applications.

JavaScript manipulates the DOM by selecting HTML elements and changing their content, attributes, or styles. For example, when a user clicks a button, JavaScript can update the text of a paragraph, display new information, or change the appearance of an element without refreshing the webpage.

A common example is an online shopping website where clicking the "Add to Cart" button immediately updates the cart count without reloading the page. Similarly, social media websites update likes and comments dynamically using DOM manipulation.

To improve performance and user experience, unnecessary DOM updates should be minimized, frequently used elements should be cached, event delegation should be used where appropriate, asynchronous data loading techniques such as AJAX or the Fetch API should be implemented, and JavaScript code should be optimized to ensure smooth and responsive webpage interactions.

index.html × # styles.css × JS script.js × + | Preview | AI Agent

```html
<p id="text">Welcome</p>

<button onclick="changeText()">Click</button>

<script>
function changeText(){
document.getElementById("text").innerHTML="Welcome to JavaScript!";
}
</script>
```

Welcome

Click